



Importance of Binary and Hexadecimal Number Systems in Computer Systems

Introduction

Binary & Hexadecimal Number Systems

Computers understand only two states – 0 and 1. These binary digits form the foundation of all data representation, storage, and processing in computer systems.

BINARY	HEXADECIMAL
10110100	0xB4
01001101	0x4D
11101010	0xEA
00011101	0x1D

→

HEX VALUES
0x1A3F
0x2B7C
0xFFEE
0x09AC

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Introduction to Number Systems

What are Number Systems?

A number system is a method of representing numbers.

Why Binary?

- Electronic circuits have only **ON (1)** and **OFF (0)** states.
- Binary is reliable for digital communication.
- Every operation inside a computer is performed using binary.

Example

Decimal 25

Binary 11001

Hexadecimal 19

Types Used in Computers

NUMBER SYSTEM	BASE	DIGITS
Decimal (Base 10)	10	0–9 0 1 2 3 4 5 6 7 8 9
Binary (Base 2)	2	0, 1 0 1
Hexadecimal (Base 16)	16	0–9, A–F 0 1 2 3 4 5 6 7 8 9 A B C D E F

10

DECIMAL
Human-friendly
number system
used in daily life.

1010

BINARY
Used by computers
internally. Represents
data using 0 and 1.

AF

HEXADECIMAL
Compact representation
of binary. Widely used in
programming and memory
addressing.



Data Representation, Storage & Processing



How Computers Represent Data

Everything inside a computer is stored as **bits (0 and 1)**.

Storage Units

- 1 Bit = 0 or 1
- 8 Bits = 1 Byte
- 1024 Bytes = 1 KB

Example

- Character **A**
- ASCII Decimal = **65**
- Binary = **01000001**
- Hex = **41**

Data Processing

CPU performs operations using binary.

5 + 3

Binary

0101

0011

1000

Result = 8



Why Hexadecimal is Important

Hexadecimal Representation

- Hexadecimal provides a shorter representation of binary.

Example

Binary and Hexadecimal Representation

Binary	Hex
10101111	AF
11111111	FF
11001010	CA

Key Point: Every 4 binary bits (a nibble) corresponds to 1 hexadecimal digit.

0 1 2 3 4 5 6 7
8 9 A B C D E F

Conversion Rule

Every 4 binary bits = 1 hexadecimal digit

Example :

1010 1111

1010 = A

1111 = F

Answer = AF



Number System Conversion & Debugging



Importance of Conversion

Binary ↔ Hexadecimal Conversion

Example :

Binary

1111000010101100

↓

1111 0000 1010 1100

↓

F0AC

Role in Debugging

Developers analyze:

- Memory dumps
- Registers
- Error codes
- Machine instructions

Example Memory Dump

Address	Data
0x1000	4A
0x1001	FF
0x1002	0C

Without hexadecimal, the same data would be much longer in binary.



Conclusion

- ✓ Binary is the fundamental language of computers.
- ✓ Hexadecimal simplifies binary representation.
- ✓ Data, instructions, and memory are represented using binary.
- ✓ Hexadecimal is widely used for:

Memory addressing

Machine instructions

Debugging

Programming



301

THANK YOU!

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